18. Perceptron-Based IRIS Classification

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.linear_model import Perceptron
from sklearn.metrics import accuracy_score

iris = load_iris()

X = iris.data
y = iris.target

X_train,X_test,y_train,y_test = train_test_split(
   X,y,test_size=0.3,random_state=42)

model = Perceptron(random_state=42)
model.fit(X_train,y_train)

p = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test,p))
print("Predicted Class:", iris.target_names[model.predict([[5.1,3.5,1.4,0.2]])[0]])
```

OUTPUT

```
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.

= RESTART: C:/Users/yuvas/Downloads/win-g2010-1_3-n_mcd/win/LIB/machine learning
 lab/ML LAB CODE.py
Accuracy: 0.4666666666666667
Predicted Class: setosa
```